

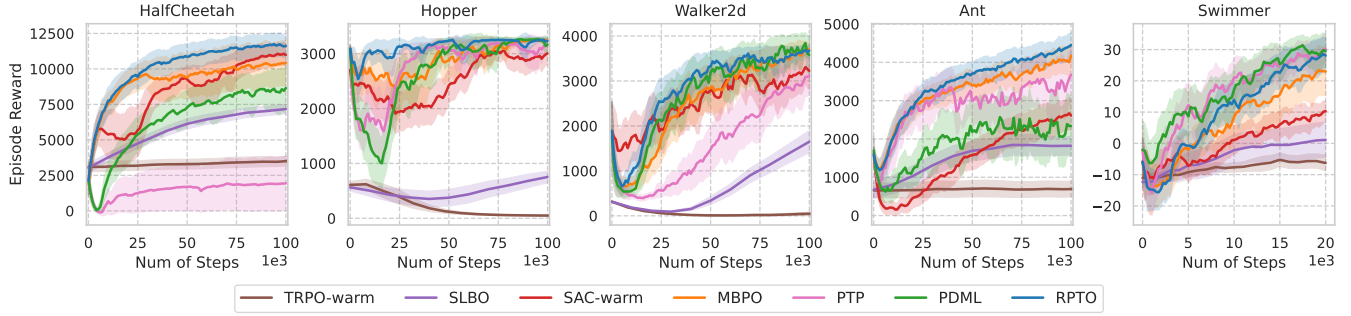


Figure 2: Overall performance on MuJoCo tasks.

trative environment to show how the proposed algorithms are practically useful. In this experiment, we evaluate the RPO, RTO and RPTO with deterministic physical dynamics model, because the physical systems are explicitly coded in CartPole, allowing us to conveniently build physical dynamics model with only a few trainable factors. In this didactic example, only the pole length is treated as the trainable parameter, i.e., ϕ in CartPole only contains one free parameter. Similarly, we also first pre-train a converged policy in the source environment, but we don't need to pre-train the dynamics model this time. We use deterministic physical dynamics to minimize the dynamics gap between the source and the target environment.

Figure 3 reports the evaluation results of our RPO, RTO and RPTO. Figure 3(a) demonstrates that RPO (without RTO) is sufficient to successfully transfer the policy to the target environment with a pole length of 1.2, and RPO transfers the policy faster than SAC-warm. In Figure 3(b), by varying the pole length to test more target environments, we find that when the pole length difference becomes larger, RPO fails to obtain a stable policy in the target environment. These results are consistent with our analysis at the RPO section. Figure 3(c) evaluates the performance of RTO. As we can observe, RTO can optimize ϕ to converge to the true pole length in the target environment. Finally, Figure 3(d) reports the RPTO's performance on the target environment with the pole length of 2.0. As we can observe, RPTO transfers the policy much faster than all the other methods.

Future Work

Our work is related to model-based RL, environment design and sim2real problems. There are some future directions worth investigating. For example, as we have discussed before, controlling the update step size or frequency of RTO can provide better curriculum learning or environment design (although in this paper we simply fix the learning rate of RTO). Connecting this with meta-RL and current environment design algorithms is a promising future direction. Moreover, the theory and algorithms in this paper are orthogonal to other techniques commonly used in sim2real tasks, such as domain adaptation, domain randomization, augmented observation, etc., and all of these methods can be integrated together into sim2real applications.

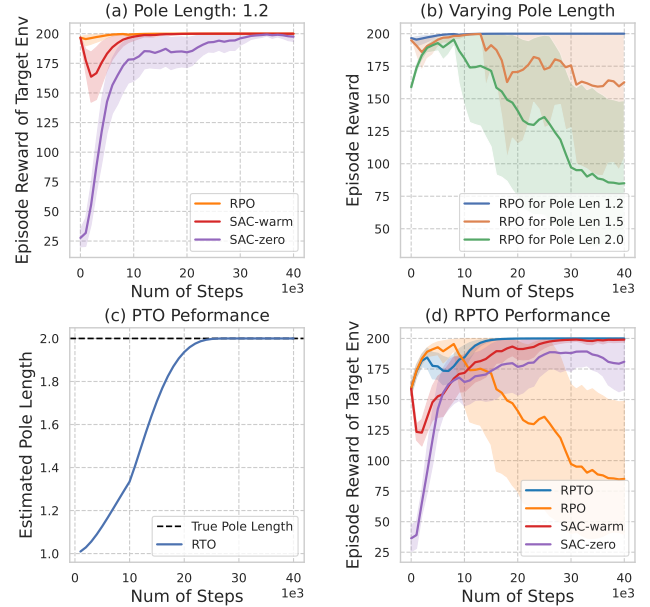


Figure 3: Illustrative experiments in CartPole. The length of pole in source environment is 1.0.

Acknowledgments

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